JR Validated Environment

Design of Experiments Guide

Planning and analysing experiments with jrc_doe_design and jrc_doe_analyse

Version 1.0

1 What Is Design of Experiments?

Design of Experiments (DoE) is a structured method for planning experiments so that you can efficiently identify which factors influence a measurement outcome and by how much. Unlike testing one factor at a time (OFAT), DoE varies multiple factors simultaneously in a structured pattern, which allows you to:

- Detect interactions between factors that OFAT would miss entirely
- Reach valid conclusions with fewer total runs
- Estimate the effect of each factor with quantified statistical confidence
- Provide documented statistical justification for design choices — essential in regulated medical device development

1.1 When to Use DoE

Situation	Recommended approach	Why
You have 2–5 factors and want to understand their effects fully	Full factorial (full2 or full3)	All main effects and interactions estimable
You have 5–8 factors and want to screen which matter	Fractional factorial or Plackett-Burman	Fewer runs; interaction information sacrificed for screening
You have 8 or more factors to screen quickly	Plackett-Burman (pb)	Minimum runs; main effects only
You need to understand curvature / non-linear response	Full factorial 3-level (full3) or add centre points to full2	Detects non-linearity without full RSM
You want to optimise a response (find best settings)	Response Surface Method (not in current scope)	RSM maps the response surface

⚠ **Important:** DoE does not replace verification testing. It is used during design characterisation and design transfer to understand process factors, not to demonstrate conformance to specification limits. For verification sample sizing, use the jrc_ss_* scripts.

2 Key Concepts

2.1 Factors, Levels, and Responses

A factor is an input variable you deliberately change during the experiment (e.g. sealing temperature, applied pressure). A level is one of the values that factor takes during the experiment. A response is the output you measure (e.g. seal strength in Newtons).

Term	Definition	Example
Factor	An input variable deliberately varied	Sealing temperature (°C)
Level	A specific value a factor takes in the design	160°C (low), 200°C (high)
Response	The measured output of the experiment	Seal strength (N)
Main effect	Change in mean response when one factor moves from low to high	Temperature effect = mean(high) − mean(low)
Interaction	Effect of one factor depends on the level of another	Temperature × Pressure
Replicate	A complete repeat of the full design	Running all 8 combinations twice = 16 runs

Centre point	A run at the midpoint of all factor ranges; detects curvature	Temperature = 180°C, Pressure = 3.5 bar

2.2 Randomisation

Run order is randomised automatically by jrc_doe_design. This distributes the effect of time-related nuisance variables (tool wear, ambient temperature drift, operator fatigue) across all experimental conditions rather than confounding them with a factor. Always run experiments in the printed randomised order. Do not re-sort the run sheet.

2.3 Coded Values

In DoE analysis, factor values are rescaled to coded units where the low level = −1 and the high level = +1 (and centre = 0 for three-level designs). Coded units make effect sizes directly comparable across factors regardless of their physical units. jrc_doe_analyse performs all modelling in coded units automatically; the output reports both coded effects and actual factor values.

3 Design Types

3.1 Full Factorial 2-Level (full2)

Tests every combination of two levels (low and high) for every factor. With k factors there are 2^k runs per replicate. All main effects and all interactions are fully estimable.

> i **Best for:** 2 to 4 factors when interactions are expected and run cost is manageable. The standard workhorse of design characterisation.

Run count (no centre points, 1 replicate): 2 factors → 4 runs, 3 factors → 8 runs, 4 factors → 16 runs, 5 factors → 32 runs.

3.2 Full Factorial 3-Level (full3)

Tests three levels (low, centre, high) for every factor. With k factors there are 3^k runs. Detects quadratic (non-linear) effects as well as main effects and interactions.

> i **Best for:** 2 to 3 factors when you suspect non-linear behaviour and need to characterise the full response surface without a dedicated RSM design.

> ⚠ **Run count:** 3-level designs grow quickly. 2 factors = 9 runs, 3 factors = 27 runs, 4 factors = 81 runs, 5 factors = 243 runs. Use only when non-linearity is expected.

3.3 Fractional Factorial 2-Level (fractional)

A carefully chosen subset of the full 2^k design. jrc_doe_design selects the minimum-run design automatically using FrF2 at the best available resolution. At Resolution III, main effects are clear of each other but aliased with two-factor interactions. At Resolution IV or higher, main effects are clear of two-factor interactions.

> i **Best for:** 5 to 8 factors when run cost is a concern and you are willing to accept some aliasing of interactions. Review the design resolution in the output summary.

3.4 Plackett-Burman Screening (pb)

An orthogonal screening design that estimates all main effects with the minimum number of runs (a multiple of 4). Interactions are not estimable. Suitable for initial screening of many factors to identify the vital few.

> **Best for:** Screening 5 or more factors when only main effects are needed. Follow up significant factors with a full factorial or fractional design.

3.5 Centre Points

Centre points are additional runs at the midpoint of all factor ranges (level = 0). They can be added to any full2 or fractional design. They serve two purposes: (1) checking for curvature in the response surface without committing to a full 3-level design, and (2) providing an independent estimate of pure error.

> **Recommendation:** Add 3 to 5 centre points to any full2 or fractional design as standard practice. They are low-cost insurance against missing non-linear effects.

3.6 Choosing a Design — DoE Planner

The JR Script Guide includes an interactive DoE Planner that walks you through three questions (number of factors, goal, run budget) and recommends the best design type with the exact jrc_doe_design command to use. Open web/script_guide.html in your browser and select the DoE Planner tab.

4 Step-by-Step Workflow

Step 1 — Define the experiment

Before touching the software, answer these questions:

- What is the response variable? (What will you measure? In what units?)
- What are the factors? (What inputs can you control?)
- What are the factor ranges? (Realistic low and high values that cover the design space of interest)
- How many runs can you afford? (This determines the design type)
- What is your null hypothesis? (e.g. Temperature has no effect on seal strength)

Step 2 — Create the factors file

Create a CSV file with one row per factor. Required columns: name, low, high. An optional mid column can specify a non-midpoint centre level; if absent, it is computed as (low + high) / 2.

```
name,low,high
Temperature,160,200
Pressure,2,5
DwellTime,3,8
```

> **Column names:** The name column becomes the factor name in the HTML run sheet and analysis output. Use short, descriptive names without spaces (e.g. SealTemp not Sealing Temperature).

Step 3 — Generate the design

Run jrc_doe_design with your chosen design type, factors file, response name, and output folder. Optionally specify centre points and replicates.

```
jrc_doe_design "full2" "~/factors.csv" "SealStrength_N" "~/experiment/" 3 1
```

Two files are written to the output folder:

- doe_design_full2_SealStrength_N_[timestamp].html — the printable run sheet. Open in your browser and use Print to produce a lab-ready sheet.
- doe_design_full2_SealStrength_N_[timestamp].csv — the data entry file. Open in a text editor or spreadsheet and fill in the response column after each run.

Step 4 — Print and run the experiment

Open the HTML file in your browser and print it. Run the experiments strictly in the printed run order (do not re-sort). Record the response value for each run in the Response column on the printed sheet.

⚠ **Critical:** Always run in the printed randomised order. Re-sorting defeats the purpose of randomisation and may introduce systematic bias that invalidates the statistical analysis.

Step 5 — Enter data

Open the companion CSV file in a text editor or spreadsheet application. Fill in the response values in the last column (do not rename any column or add rows). Save the file.

ℹ **Note:** The first line of the CSV is a comment starting with #. Do not delete or edit this line — jrc_doe_analyse uses it to identify the design type and response name.

Step 6 — Analyse the results

Run jrc_doe_analyse with the completed CSV file and an output folder.

```
jrc_doe_analyse "~/experiment/doe_design_full2_SealStrength_N_[timestamp].csv"
"~/experiment/"
```

The analysis report (HTML file) is written to the output folder. Open it in your browser.

Step 7 — Interpret and document

Review the Pareto chart, ANOVA table, and main effects / interaction plots in the analysis report. See Section 6 for interpretation guidance. Document your findings per Section 7.

5 Worked Example: Sealing Process Characterisation

A medical device team is characterising a heat-sealing process. Three factors are believed to influence seal strength: sealing temperature, applied pressure, and dwell time. The team runs a full factorial 2-level design with 3 centre points.

5.1 Experiment Setup

Factor	Symbol	Low (−1)	High (+1)
Sealing temperature	Temperature	160 °C	200 °C
Applied pressure	Pressure	2 bar	5 bar

Dwell time	DwellTime	3 s	8 s

Response: SealStrength_N (seal peel strength in Newtons). Target: > 15 N.

Design: full2 with 3 centre points, 1 replicate. Total runs: 8 + 3 = 11.

5.2 Factors File

```
name,low,high
Temperature,160,200
Pressure,2,5
DwellTime,3,8
jrc_doe_design "full2" "~/sealing/factors.csv" "SealStrength_N" "~/sealing/" 3
1
```

5.3 Results (After Running the Experiment)

After completing all 11 runs and filling in the companion CSV, the team observes the following pattern:

- High temperature consistently produces higher seal strength
- High pressure also tends to increase strength
- Dwell time appears to have a smaller effect
- The interaction between Temperature and Pressure is visible — at high pressure, increasing temperature has a larger effect

5.4 Analysis

```
jrc_doe_analyse "~/sealing/doe_design_full2_SealStrength_N_[timestamp].csv"
"~/sealing/"
```

Typical analysis output for this experiment:

Term	Sum Sq	F value	p-value
Temperature	84.5	31.2	0.003
Pressure	38.1	14.1	0.019
DwellTime	4.2	1.6	0.277
Temperature:Pressure	22.8	8.4	0.044
Temperature:DwellTime	2.1	0.8	0.422
Pressure:DwellTime	1.4	0.5	0.516
Residuals	8.1		

Curvature test: not significant ($p = 0.31$). No evidence of non-linearity — the 2-level model adequately describes the data.

5.5 Interpretation

The Pareto chart confirms that Temperature has the largest effect, followed by Pressure and the Temperature × Pressure interaction. DwellTime and its interactions are not significant at $\alpha = 0.05$.

The interaction between Temperature and Pressure means the optimal settings depend on both factors simultaneously. At high pressure (5 bar), increasing temperature from 160°C to 200°C increases seal strength by approximately 12 N. At low pressure (2 bar), the same temperature increase produces only a 6 N gain.

Design conclusion: Set sealing temperature to 200°C and pressure to 5 bar. Dwell time can be set to the lower value (3 s) without loss of seal strength, reducing cycle time. Document this conclusion with the analysis report HTML as evidence.

6 Interpreting the Analysis Report

6.1 ANOVA Table

The ANOVA table partitions the total variation in the response into contributions from each model term. Key columns:

Column	Meaning	What to look for
Sum Sq	Variance attributed to this term	Larger values indicate more influential terms
F value	Ratio of term mean square to residual mean square	F > 1 means the term explains more variance than noise
Pr(>F)	p-value: probability of this F value by chance alone	$p < 0.05$: significant at $\alpha = 0.05$ (highlighted in blue)

Note: Significance depends on residual degrees of freedom. Unreplicated designs with many factors may have few residual df, making it difficult to achieve significance. Consider adding replicates or reducing the number of terms in the model.

6.2 Pareto Chart of Standardised Effects

The Pareto chart ranks all model terms by their absolute standardised effect (effect size divided by the residual standard error). The vertical dashed line marks the threshold for significance at $\alpha = 0.05$ ($|\text{effect}| = 2.0$). Bars crossing the line are significant.

The Pareto chart is a quick visual scan: if the longest bar is well to the right of the line, that factor dominates. If all bars are clustered near zero, the response may not be sensitive to any of the tested factors — or the design has insufficient power.

6.3 Main Effects Plot

The main effects plot shows the mean response at the low and high level of each factor (across all other factor combinations). A steep slope indicates a large main effect. A flat line indicates the factor has little influence on the response.

Caution: Do not use the main effects plot alone if significant interactions are present. An interaction means the effect of one factor is not constant across levels of another — the main effects plot averages over this and can be misleading.

6.4 Interaction Plot

The interaction plot shows mean response for all combinations of two factors. Two parallel lines indicate no interaction. Non-parallel or crossing lines indicate an interaction.

Interaction plots are generated for all factor pairs in full2 and fractional designs with $k \leq 6$ factors. For larger designs or Plackett-Burman, interactions are not estimable and this section is omitted.

6.5 Curvature Test

If centre points were included in the design, the analysis report includes a curvature test. This compares the mean response at the centre points with the mean response predicted by the linear model (the average of the factorial points).

- ✅ Not significant ($p \geq 0.05$): the linear model is adequate. No evidence of quadratic effects.
- ⚠️ Significant ($p < 0.05$): curvature is present. The linear + interaction model does not fully describe the response surface. Consider upgrading to a 3-level (full3) design or adding axial points for a Response Surface Method analysis.

7 FDA and ISO 13485 Documentation

7.1 What to Record

For a DoE study conducted as part of design characterisation or design transfer, document the following in your Design History File (DHF) or Device Master Record (DMR):

- Objective and rationale: why was DoE used rather than OFAT? What design choice was made and why?
- Factors and levels: factors CSV file and justification for level selection (based on design specifications or process knowledge)
- Design type and run count: type, number of runs, randomisation seed (from the HTML run sheet)
- Raw data: completed companion CSV file with all run results
- Analysis report: HTML output from jrc_doe_analyse (attach or reference by filename including timestamp)
- Conclusions: which factors are significant, what settings are selected, and why
- Limitations: any deviations from the run order, outliers excluded, or model assumptions not met

7.2 Relating DoE to Design Controls

Under FDA 21 CFR Part 820.30 (Design Controls) and ISO 13485 Section 7.3, design verification and validation activities must be planned and documented. A DoE study may serve as:

- Design verification evidence (demonstrating that design outputs meet design inputs) if the response is a specified performance parameter
- Design transfer evidence (demonstrating that the manufacturing process consistently produces a conforming device) when factors are process parameters
- Risk management support (ISO 14971): identifying which factors have the largest effect size informs risk controls and process monitoring requirements

⚠️ **Important:** DoE analysis establishes statistical relationships between factors and a response under the conditions tested. Extrapolation outside the tested factor ranges is not statistically justified. Document the factor ranges explicitly.

7.3 Version Control

The HTML run sheet and analysis report both include a generation timestamp and randomisation seed. These serve as traceability links between the raw data and the analysis. Store both files in your document management system with version control.

8 Glossary

Term	Definition
ANOVA	Analysis of Variance. A statistical method that partitions total response variation into contributions from each model term.
Aliasing	In fractional designs, two or more effects cannot be estimated independently — they are confounded with each other.
Centre point	An experimental run at the midpoint of all factor ranges. Used to detect curvature and estimate pure error.
Coded value	A rescaled factor value where low = −1, centre = 0, and high = +1. Used internally by jrc_doe_analyse for comparable effect sizes.
Confounding	See Aliasing.
Curvature	A non-linear (quadratic) relationship between a factor and the response. Detected by comparing centre point responses to factorial point predictions.
DoE	Design of Experiments.
Factor	An input variable deliberately changed during an experiment.
Fractional factorial	A subset of the full factorial design that estimates main effects and some interactions with fewer runs.
Interaction	The effect of one factor on the response depends on the level of another factor.
Level	A specific value of a factor in the experimental design.
Main effect	The average change in response when a factor moves from its low to its high level.
OFAT	One Factor At a Time. A less efficient alternative to DoE that misses interactions.
Pareto chart	A bar chart ranking effects by size. Bars crossing the significance threshold (|effect| = 2.0) are statistically significant.
Plackett-Burman	An efficient screening design for many factors using only a multiple-of-4 run count. Main effects only.
Randomisation	Running experiments in a random order to distribute the effect of time-related nuisance variables.
Replicate	A complete repeat of the design. Increases power and provides an estimate of pure error.
Resolution	A measure of aliasing in fractional designs. Resolution III: main effects aliased with 2-factor interactions. Resolution IV: main effects clear of 2-factor interactions. Resolution V: 2-factor interactions clear of each other.
Response	The measured output of the experiment.
Standardised effect	Effect size divided by residual standard error. Used in the Pareto chart for comparability across factors.